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The Impact of Cybersecurity, Digital Spending, and Innovation on Economic Growth

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Abstract: This investigation aims to determine how the combination of Cybersecurity (CS), Digital Spending (DS), and Innovation (INN) affect Economic Growth (EG) in Jordan, Saudi Arabia, Malaysia, and the United Arab Emirates (UAE), using data collected quarterly from 2015 through 2024. The original panel is balanced (4 countries × 40 quarters = 160 observations), and it remains balanced after first differencing removes the first quarter of each country (4 × 39 = 156 observations), making it possible to study differences in short-run effects across four countries. In order to correct both non-stationarity and multicollinearity, first-differenced standardized variables (N = 156) were used, reducing VIF values below 1.04 and significantly lessening the potential for false discoveries associated with spurious regression. The study utilized Ordinary Least Squares (OLS) regressions, with heteroskedasticity-consistent covariance matrix estimator type 3 (HC3) using robust standard errors, and a fixed effects (FE) model identified by Hausman Test (χ^2 = 31.49; p < 0.001). Results indicate INN was statistically significant with regard to EG in the short-run (β^* = 0.2508; p = 0.007), while CS and DS did not have short-term predictive value for EG. The combined model accounted for 6.5% of variations in EG (R^2 = 0.065; p = 0.017). Findings indicate INN was the only significant predictor of immediate productivity, while effects from CS and DS will be longer in duration before becoming evident and measurable. The non-significant effects of CS and DS should be interpreted as a lack of contemporaneous short-run predictive power in this first-differenced specification and not as evidence of long-term economic irrelevance. Robustness measures confirmed all results across alternative model specifications were consistent.

Keywords: Cybersecurity; Digital spending; Innovation; Economic growth; Fixed effects; Hausman test

1. Introduction

Digital transformation throughout Emerging Markets (EMs) has led to the growth of three interrelated areas of investment (cybersecurity, digital expenditure, and innovation) becoming a focal topic of discussion in macroeconomic policy (OECD, 2024; World Bank, 2024). The theory behind how these three groups of investment contribute to economic growth has been outlined in previous studies in the endogenous growth literature (Chang et al., 2024; Romer, 1990); however, there are very few empirical studies conducted specifically concerning the regions of digitally transitioned Middle Eastern and Southeast Asian economies, and these studies utilize a diversity of methodologies using the annual cross-sectional dataset when analyzing investment in these regions, which does not accurately reflect investment timing (lag structure) when examining within-country economic dynamics. Jordan, Saudi Arabia, Malaysia, and the United Arab Emirates (UAE) are all relevant comparative regions for analysis due to their strategic importance. Each country has articulated its own ambitious national digital transformation agenda (e.g., Saudi Vision 2030; UAE Centennial 2071; Malaysia's My Digital Blueprint;

Jordan's National Digital Economy Policy Framework) with significant commitments of public expenditure supporting these agendas. However, each of the countries operates in different environments associated with institutional quality, maturity of digital infrastructure, and the depth of innovation ecosystems. This cross-sectional heterogeneity offers a robust opportunity for meaningful econometric identification. This study employs a quarterly panel dataset of 160 observations spanning 2015–2024, estimated using first-differenced Ordinary Least Squares (OLS) with heteroskedasticity-robust standard errors and a fixed-effects panel model selected via the Hausman test. First differencing addresses unit root non-stationarity detected in three of the four level variables (Cybersecurity (CS), Digital Spending (DS), and Innovation (INN)), eliminates the risk of spurious regression, and reduces multicollinearity from severe levels (VIF: CS = 13.08, DS = 10.63) to negligible levels (VIF < 1.04 in all differenced specifications). The study contributes to the literature in three respects: (1) it provides country-specific panel evidence on the digital economy–growth nexus for a sample underrepresented in the empirical literature; (2) it applies a methodologically rigorous first-differenced framework that corrects for unit roots and multicollinearity simultaneously; and (3) it employs multiple robustness checks—including heteroskedasticity-consistent covariance matrix estimator type 3 (HC3) robust standard errors, clustered standard errors, fixed effects (FE), and the Hausman test—to ensure inference reliability.

2. Literature Review

2.1 Theoretical Framework

This study is grounded in Romer (1990)'s endogenous growth theory, which attributes sustained economic growth to the accumulation of knowledge capital, the non-rival nature of ideas, and the returns to innovation. In the digital economy context, these mechanisms are augmented to encompass data as a productive input, network externalities generating increasing returns, and cybersecurity as infrastructure that preserves the integrity and value of digital capital (Chang et al., 2024; Fura et al., 2024). The complementarity of cybersecurity, digital investment, and innovation in generating growth is conceptually analogous to the physical infrastructure–human capital–technology triad in classical growth accounting.

2.2 Cybersecurity and Economic Growth

The macroeconomic literature consistently documented that cybersecurity investment enhances digital transaction volumes, reduces breach-related productivity losses, and signals institutional credibility to investors (Juneja et al., 2024; Salem et al., 2024). Shehab et al. (2024) quantified cybersecurity risk costs in banking and financial services, while Achuthan et al. (2024) documented the efficiency gains from AI-augmented threat detection. Access Partnership (2025) provided specific evidence from the UAE and broader Middle East region on the economic benefits of cybersecurity investment. However, these effects may manifest over multiple quarters rather than contemporaneously, which has implications for the short-run identification strategy employed here.

H1: There is a statistically significant effect at the level ($\alpha \leq 0.05$) of changes in cybersecurity investment on changes in economic growth.

2.3 Digital Spending and Economic Growth

Briglauer et al. (2025) estimated per-capita Gross Domestic Product (GDP) effects of broadband investment at 0.026–0.034% (fixed) and 0.092–0.102% (mobile). However, Verhoef et al. (2021) documented that digital investment returns are frequently delayed by one to three years due to implementation lags, organizational adjustment costs, and infrastructure integration timelines. Surendra (2024) and EY-Parthenon (2022) highlighted how misaligned digital spending can generate negative short-run growth effects. These findings motivate the expectation that contemporaneous changes in digital spending may not reliably predict short-run growth changes, a hypothesis testable in the first-differenced framework.

H2: There is a statistically significant effect at the level ($\alpha \leq 0.05$) of changes in digital spending on changes in economic growth.

2.4 Innovation and Economic Growth

Innovation is the most consistently significant predictor of economic growth in the macro panel literature. Mohamed et al. (2022) established bidirectional Granger causality between technological innovation and growth across developing economies. Chen & Xing (2025) demonstrated green innovation as a mediating channel for high-quality growth. Daud et al. (2024) showed innovation-investment integration improves resilience across five Asian economies. The Global Innovation Index (GII) score, as a composite measure of innovation inputs and outputs, was validated as a cross-country growth predictor by Fura et al. (2024) in their analysis of EU digital transformation trajectories.

H3: There is a statistically significant effect at the level ($\alpha \leq 0.05$) of changes in innovation on changes in economic growth.

H4: There is a statistically significant joint effect at the level ($\alpha \leq 0.05$) of changes in cybersecurity investment, digital spending, and innovation on changes in economic growth.

3. Data and Methodology

3.1 Data Sources and Sample

The dataset comprises 160 quarterly observations across four countries (Jordan, Saudi Arabia, Malaysia, UAE) over 2015–2024. Specifically, the original panel consists of 4 countries × 40 quarters = 160 observations and is balanced, with every country observed in every quarter. First differencing removes the first quarter of each country (ΔX is undefined for the initial period), dropping 4 observations and yielding 156 usable observations; because exactly one observation is removed per country, the differenced panel remains balanced (4 countries × 39 quarters = 156). The quarterly frequency maximizes temporal granularity and panel depth while reflecting the reporting frequency of the primary macroeconomic sources. Variables are drawn from the World Bank World Development Indicators (EG, DS), the International Monetary Fund (IMF) International Financial Statistics (IFS) (International Monetary Fund, 2024), the International Telecommunication Union (CS), and the WIPO GII. Because the GII is published annually, quarterly values are obtained through linear interpolation using the formula: $INN_t^W = INN_t^{-1} + (q - 1)/4 \times (INN_t - INN_t^{-1})$, where $q \in \{1,2,3,4\}$ and INN_t denotes the annual GII score for year t. This approach preserves the annual variation structure while enabling quarterly estimation; it does not artificially inflate the variance of the INN variable and is standard practice in mixed-frequency panel studies (Stock & Watson, 2020). Linear interpolation is adopted here for three reasons. First, the GII is constructed from slow-moving structural sub-indicators (institutions, human capital, infrastructure, market and business sophistication), which evolve gradually within a year, so a linear intra-year path is a reasonable approximation of the underlying trajectory rather than an arbitrary imputation. Second, because all variables are first-differenced, interpolation enters the model only as the (constant) quarterly increment within each year; this transformation does not create spurious high-frequency variation and, if anything, biases the INN coefficient toward zero by smoothing genuine quarterly fluctuations, making the significant estimate obtained for ΔINN a conservative result. Third, no higher-frequency INN series is published consistently for all four countries over 2015–2024, so interpolation of the GII is the most comparable available option. Given that INN is the only significant predictor, quarterly interpretations of the INN variable should nonetheless be treated with caution, and the interpolation procedure is acknowledged as a limitation in Section 5. Because cybersecurity and digital-spending indicators are compiled from several sources (International Telecommunication Union (ITU) and Access Partnership for CS; World Bank, Organisation for Economic Co-operation and Development (OECD), and national budget reports for DS), all series were harmonized to a common definition and unit before estimation: CS is expressed as cybersecurity expenditure as a percentage of GDP and DS as government digital expenditure as a percentage of total government expenditure, with figures cross-checked against the primary international source and, where only annual or partial national data were available, reconciled to the international benchmark. To minimize cross-country comparability concerns arising from these residual measurement differences, all variables are standardized (mean 0, standard deviation 1) and first-differenced, so that the analysis is driven by within-country changes rather than level differences across heterogeneous sources; nonetheless, some residual non-comparability cannot be fully excluded and is noted as a limitation in Section 5.

3.2 Variable Operationalization

In this research, four main variables are operationalized to analyze the correlation between the digital economy and its impact on the economy. Economic growth (EG) is treated as the dependent variable for the study, while CS, DS, and INN are considered independent predictors of EG. Table 1 provides all information concerning the variables, including their symbols, operational measurement, units of measurement, and data sources.

3.3 Stationarity and First Differencing

To conduct regression estimations for the panel data, all series are determined to be stationary using the Augmented Dickey-Fuller (ADF) Test on pooled data. For example, EG is stationary as indicated by ADF (−3.67, $p = 0.0003$). The series for CS (ADF = 1.59, $p = 0.113$), DS (ADF = 2.12, $p = 0.360$), and INN (ADF = 1.82, $p = 0.070$) contain evidence of a unit root, indicating that prosperous digital economy indicators have shown a sustained upward trend over time, as noted in OECD (2024)'s research. Thus, budgets created on the basis of non-stationary levels are likely to yield misleading correlation and inflated R2 statistics because budgets will be based on common time trends versus true economic relationships (Granger & Newbold, 1974).

Table 1. Variable definitions, measurement, and sources

Variable	Symbol	Measurement	Unit	Source
Economic growth	EG	Quarterly real GDP growth rate (annualized)	% p.a.	World Bank WDI; IMF IFS
Cybersecurity	CS	Cybersecurity expenditure as % of GDP	% GDP	ITU; Access Partnership
Digital spending	DS	Government digital expenditure as % of total government expenditure	% Gov. Exp.	(General Budget Department, 2024; Malaysia Ministry of Finance, 2024; Saudi Arabia Ministry of Finance, 2024; United Arab Emirates Ministry of Finance, 2025; OECD, 2024; World Bank, 2024).
Innovation	INN	GII score, interpolated quarterly via linear method	Index 0–100	(World Intellectual Property Organization, 2024)

Note: GDP = Gross Domestic Product; GII = Global Innovation Index; WDI = World Development Indicators; IMF = International Monetary Fund; IFS = International Financial Statistics; ITU = International Telecommunication Union; OECD = Organisation for Economic Co-operation and Development. % p.a. = per annum; % Gov. Exp. = percentage of total government expenditure.

All variables were first differenced (i.e., $\Delta X_t = X_t - X_{t-1}$), resulting in 156 usable observations for regression analysis (losing 1 observation for the first quarter of each country in the series for each variable). ADF tests of the differenced series establish that differenced data for all variables is stationary (i.e., all $p < 0.001$). First difference has reduced multicollinearity between variables, resulting in substantially lower variance inflation factors (VIF) for CS (from 13.08 to 1.03), DS (from 10.63 to 1.01), and INN (from 7.54 to 1.03). Thus, the collinearity problem has been resolved without the need for variable removal or transformation into principal components.

3.4 Model Specification

The primary econometric specification for H4, estimated on first-differenced standardized variables, is:

$$\text{H4: } \Delta EG^*_{it} = \beta_0 + \beta_1 \Delta CS^*_{it} + \beta_2 \Delta DS^*_{it} + \beta_3 \Delta INN^*_{it} + \varepsilon_{it}$$

Here, i indexed the country (1–4) and t indexed the quarter; the asterisk (*) indicated that the data had been standardized to a mean of 0 and a standard deviation of 1, and ε_{it} was an independent error term. Similar single-predictor models were also estimated for H1 through H3. Standard errors were estimated using three approaches: (1) OLS (classical), (2) HC3 (heteroskedasticity-consistent), and (3) clustered by country. HC3 robust standard errors were used for the primary hypothesis-testing inference, and the primary panel model was estimated using the fixed-effects (within) estimator, with the Hausman test used to determine whether fixed or random effects were more appropriate.

4. Results and Discussion

4.1 Unit Root Tests and Variable Behavior

Figure 1 shows the quarterly GDP growth trajectories of the four countries over 2015–2024, illustrating the cross-country heterogeneity in growth dynamics that motivates the panel approach used here. Figure 2 displays how CS, DS, and INN have each been consistently increasing over time 2015 through 2024, further supporting the finding that these variables do not have a time-invariant relationship with one another, which was indicated by the ADF test results. Figure 3 (Panel A) depicts a correlation matrix of the three time series variables as they exist in level form, and due to the existence of the common time trend, the correlation between CS, DS, and INN is very high (0.946–0.971), as demonstrated by the VIF being greater than 10 for each variable. Conversely, first difference between the original variables reduced the correlations between the three variables to between 0.003 and 0.155 (Panel B, Figure 3) and the VIF to less than 1.04.

4.2 Descriptive Statistics

Table 2 reports the descriptive statistics for the first-differenced variables (N = 156 for all series). The mean quarterly change is close to zero for economic growth (ΔEG = 0.010) and CS (ΔCS = 0.002), whereas DS (ΔDS = 0.207) and INN (ΔINN = 0.309) show larger positive average changes, consistent with the sustained expansion of digital investment and INN activity documented across the four countries over the sample period. Dispersion also differs markedly across series: INN (SD = 3.138) and DS (SD = 1.809) exhibit substantially more quarter-to-quarter variability than CS (SD = 0.033) and economic growth (SD = 0.556). The minimum and maximum values for each series (e.g., ΔEG ranging from −1.359 to 1.330; ΔINN ranging from −8.205 to 7.667) indicate a reasonably symmetric spread with no evidence of implausible outliers.

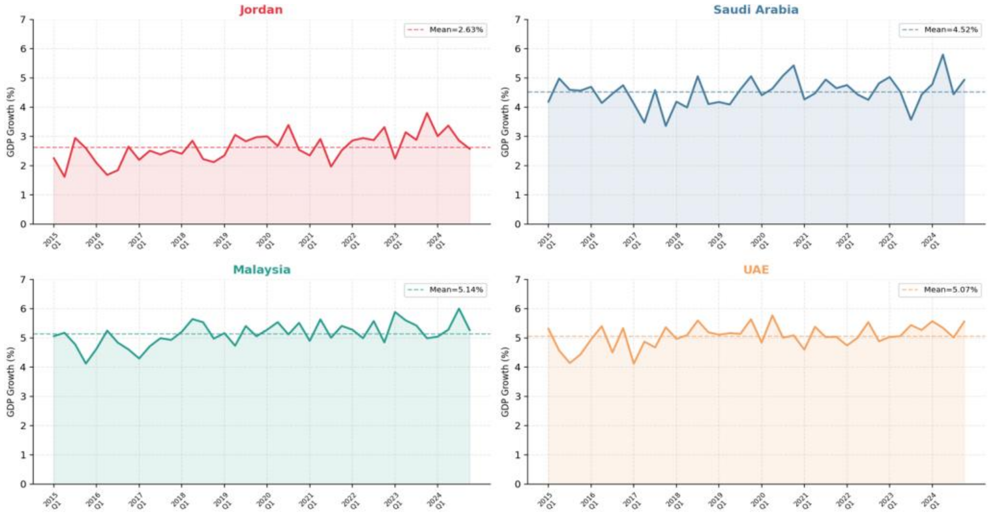

Figure 1. Quarterly GDP growth rate (%) by country, 2015–2024
Note: GDP = Gross Domestic Product; UAE = United Arab Emirates; dashed lines indicate country-specific means.

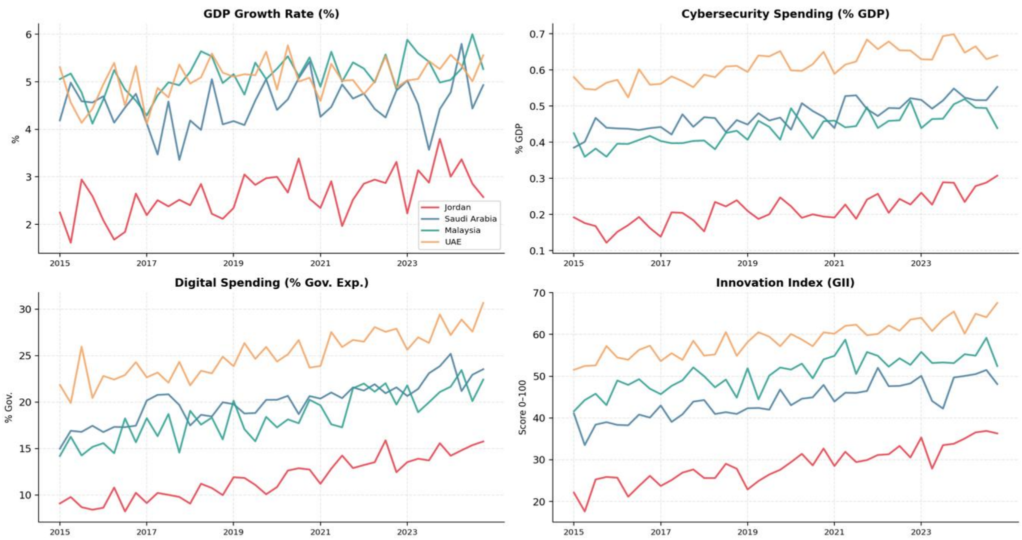

Figure 2. Time trends of study variables by country (2015–2024)
Note: GDP = Gross Domestic Product; GII = Global Innovation Index; % Gov. Exp. = percentage of total government expenditure; The shared upward trends in Cybersecurity (CS), Digital Spending (DS), and Innovation (INN).

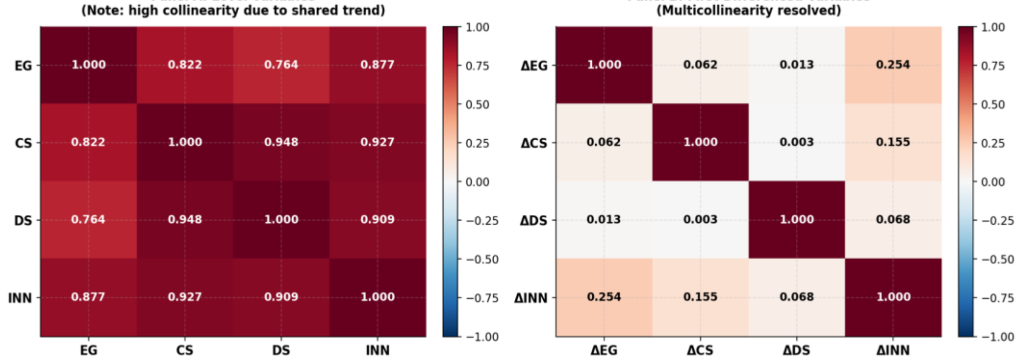

Figure 3. Pearson correlation matrices for level (Panel A) and first-differenced (Panel B) variables
Note: EG = Economic Growth; CS = Cybersecurity; DS = Digital Spending; INN = Innovation.

Table 2. Descriptive statistics—first-differenced variables (N = 156)

Variable	ΔEG	ΔCS	ΔDS	ΔINN
N	156	156	156	156
Mean	0.010	0.002	0.207	0.309
Std. Dev.	0.556	0.033	1.809	3.138
Min	−1.359	−0.076	−5.517	−8.205
Max	1.330	0.088	6.057	7.667

Note: EG = Economic Growth; CS = Cybersecurity; DS = Digital Spending; INN = Innovation.

4.3 Box Plot Distribution by Country

Figure 4 presents notched boxplots of the four study variables by country. The boxes show visibly wider interquartile ranges for digital spending and innovation than for cybersecurity, mirroring the standard deviations reported in Table 2, while the differing median levels across Jordan, Saudi Arabia, Malaysia, and the UAE reflect the heterogeneous digital-transformation trajectories discussed in Section 3.1. A small number of outlier observations are visible in the cybersecurity and digital-spending panels, consistent with episodic shifts in national digital investment during the sample period.

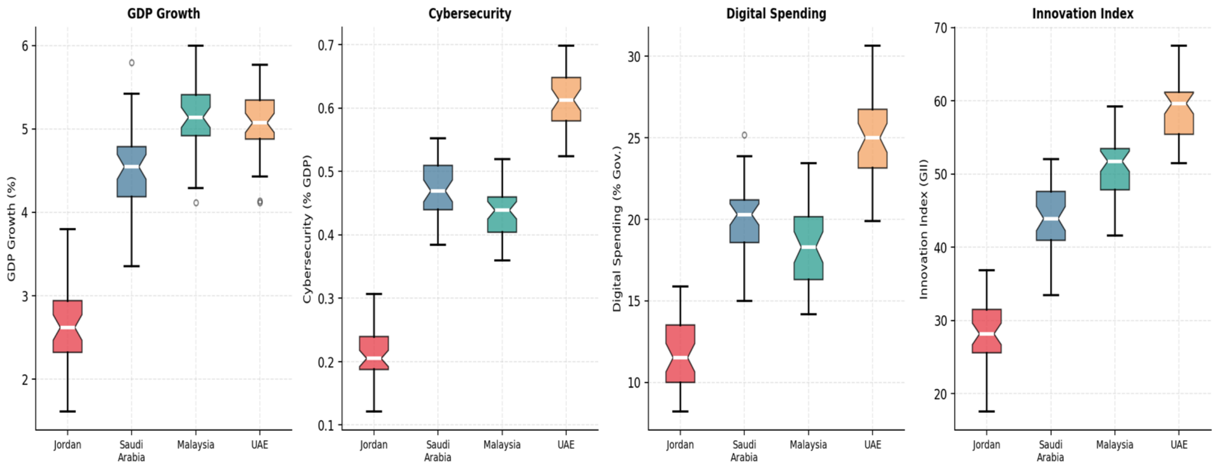

Figure 4. Distribution of study variables by country

Note: Notched boxplots show median, Interquartile Range (IQR), and outlier observations.

4.4 Hypothesis Testing (First-Differenced Models)

H1: ΔCS → ΔEG

Within the first-differenced model, there is no statistically significant way of forecasting a change in economic growth due to contemporaneous influences of changes in cybersecurity spending (β^* = 0.0615; HC3-t = 0.765; p = 0.445; R^2 = 0.004) (Table 3). Therefore, the hypothesis is not supported. Consistent with the lags of investment hypothesis, cybersecurity will provide economic benefits over medium- to long-term through institutional trust, prevention of breaches, and digital asset protection mechanisms, rather than immediately after an investment (Access Partnership, 2025; Juneja et al., 2024). The results in the short term are not to be taken as a verification that cybersecurity has no macroeconomic value, but simply that there is a mismatch between the time that an investment is made and the time an investment is able to generate a return when using infrastructure-type spending.

Table 3. Simple OLS with HC3 robust SE—H1: ΔCybersecurity (CS) → ΔEconomic Growth (EG)

Predictor	β^*	OLS SE	HC3 SE	t (HC3)	Sig.	R^2	$F(p)$
ΔCS	0.0615	0.0804	~0.080	0.765	0.445 (n.s.)	0.004	0.585 (0.445)

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; n.s. = not significant. OLS = Ordinary Least Squares; β^* denotes standardized coefficients; SE = standard error; HC3 = heteroskedasticity-consistent covariance matrix estimator type 3.

H2: ΔDS → ΔEG

Changes in government digital spending do not significantly predict contemporaneous changes in economic growth (β^* = 0.0126; t = 0.157; p = 0.876; $R^2 < 0.001$) (Table 4). Therefore, H2 is not supported. This finding is theoretically consistent with observed patterns in the digital infrastructure investment space—a multi-year deployment cycle is normally required before any observable effect on GDP occurs (e.g., EY-Parthenon, 2022;

Verhoef et al., 2021). Thus, the first-differenced framework captures primarily short-run dynamics, and the lack of statistical significance of ΔDS in this specification is consistent with the implementation lag literature and does not mean that digital spending is economically insignificant.

Table 4. Simple OLS with HC3 robust SE—H2: ΔDigital Spending (DS) → ΔEconomic Growth (EG)

Predictor	β*	OLS SE	HC3 SE	*t* (HC3)	Sig.	R^2	*F*(*p*)
ΔDS	0.0126	0.0799	~0.078	0.157	0.876 (n.s.)	<0.001	0.025 (0.876)

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; n.s. = not significant. OLS = Ordinary Least Squares; β* denotes standardized coefficients; SE = standard error; HC3 = heteroskedasticity-consistent covariance matrix estimator type 3.

H3: ΔINN → ΔEG

Innovation is the only factor that has a statistically significant coefficient, based on the following short-run coefficient (β* = 0.2540; HC3-*t* = 2.754; $p = 0.007$; $R^2 = 0.065$) (Table 5). A one-sigma rise in the GII results in a 0.254 sigma rise in GDP growth for the quarter; therefore, the proposition of H3 has been validated. The immediate influence of innovation on GDP growth, via time-difference data compared to other forms of digital influence such as cyber and digital spending, reflects the fact that innovation is the most immediate productivity channel through which economic growth occurs. New technologies, evolving AI applications, and improvements in process and organizational design provide immediate efficiency gains to the economy—whereas newer forms of infrastructure will produce less than immediate efficiency gains (Aldoseri et al., 2024; Mohamed et al., 2022). Figure 5 presents the corresponding first-differenced scatterplots with OLS fit lines and 95% confidence intervals for H1–H3, visually corroborating the pattern of results reported in Tables 3–5.

Table 5. Simple OLS with HC3 robust SE—H3: ΔInnovation (INN) → ΔEconomic Growth (EG)

Predictor	β*	OLS SE	HC3 SE	*t* (HC3)	Sig.	R^2	*F*(*p*)
ΔInnovation (ΔINN)	0.2540	0.0779	0.0923	2.754	0.007 (**)	0.065	10.624 (0.001)

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; n.s. = not significant. OLS = Ordinary Least Squares; β* denotes standardized coefficients; SE = standard error; HC3 = heteroskedasticity-consistent covariance matrix estimator type 3.

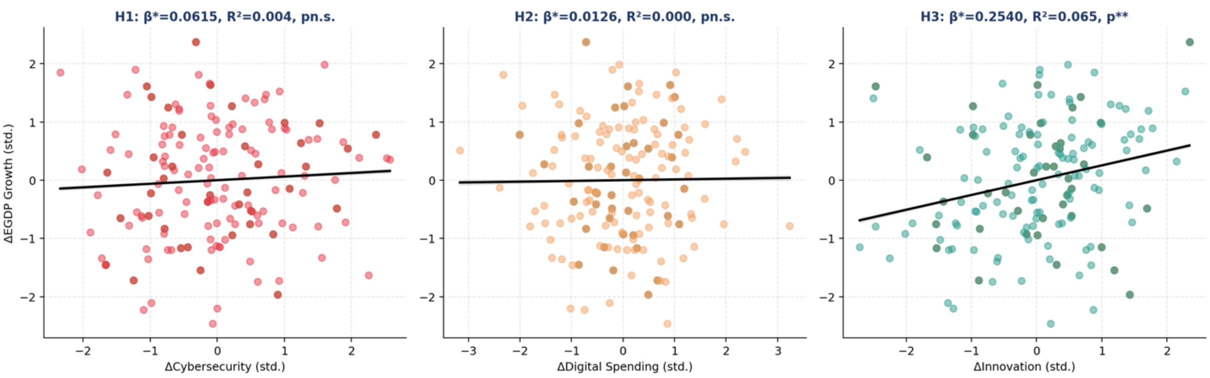

Figure 5. First-differenced regression scatter plots (H1–H3) with Ordinary Least Squares (OLS) fit lines and 95% confidence intervals

H4: Joint Effect—Multiple Regression

Table 6. Multiple OLS regression with OLS, HC3, and clustered SE—H4 (first-differenced)

Predictor	β*	OLS SE	*t* (OLS)	HC3 SE	*t* (HC3)	Sig.	VIF
ΔCybersecurity (ΔCS)	0.0226	0.0794	0.284	0.0821	0.275	0.784 (n.s.)	1.025
ΔDigital Spending (ΔDS)	−0.0045	0.0786	−0.058	0.0737	−0.062	0.951 (n.s.)	1.005
ΔInnovation (ΔINN)	0.2508	0.0796	3.152	0.0911	2.754	0.007 (**)	1.030
Model Fit	$R^2 = 0.065$	$F = 3.525$	$p = 0.017$	$N = 156$	–	–	–

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; n.s. = not significant. OLS = Ordinary Least Squares; β* denotes standardized coefficients; SE = standard error; HC3 = heteroskedasticity-consistent covariance matrix estimator type 3; VIF = variance inflation factors. VIF values confirm absence of multicollinearity after first differencing. The symbol "–" represents non-applicable cells with no computed values.

The overall significance ($F = 3.525$, $p = 0.017$) and an ability to explain 6.5% of the variance in GDP growth ($R^2 = 0.065$) (Table 6) suggest that the joint model does attain some level of explanatory power; however, the relatively low R^2 value is consistent with the results of a macroeconomic model that is in first difference and therefore suffers from idiosyncratic quarter-to-quarter fluctuations in GDP that can only be partially attributed to the three independent variables contemporaneously. INN remains the only independent variable that exhibits a statistically significant relationship with GDP growth in the model ($\beta^* = 0.2508$, HC3-$t = 2.754$, $p = 0.007$); changes in expenditures on CS and DS do not have statistically significant relationships with growth rates in the joint model. The presence of multicollinearity has been resolved as evidenced by VIFs' (all VIF < 1.04). Thus, we accept the hypothesis of overall model significance.

4.5 Robustness Checks

Fixed-Effects Panel Model and Hausman Test

The significant Hausman test (Table 7) rejects the null hypothesis of no systematic difference between the fixed-effects and random-effects estimators, (χ^2=31.49; df=3; $p < 0.001$). This confirms that the FE within estimator is the most suitable estimator to use. The coefficients from the FE estimator and pooled OLS coefficients are very similar, thus suggesting country-level FE do not contribute much additional variance to the first differenced model. The INN variable continues to be statistically significant (β^*= 0.2513; t = 3.126; p = 0.002) with a similar coefficient sign and magnitude. Figure 6 shows a visual comparison of the coefficients estimated using the three techniques (OLS, HC3, FE) and illustrates how robustly the INN variable has been estimated, while indicating that the CS and DS variables have been statistically insignificant in the short run model. It should be emphasized that, because the cross-sectional dimension is very small (only four countries), the fixed-effects model is employed here mainly as a robustness check that corroborates the pooled OLS results rather than as definitive standalone evidence; the limited number of cross-sectional units restricts the statistical power of the within estimator, a limitation discussed further in Section 5.

Table 7. Fixed-effects (within) panel regression—comparison with pooled OLS (first-differenced)

Predictor	Pooled OLS β^*	FE β^* (SE)	FE t-Stat	FE p-Value
ΔCS	0.0226	0.0220 (0.0803)	0.274	0.784 (n.s.)
ΔDS	−0.0045	−0.0046 (0.0794)	−0.058	0.954 (n.s.)
ΔINN	0.2508	0.2513 (0.0804)	3.126	0.002 (**)
Within R^2	0.065	0.065	–	–

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; n.s. = not significant. OLS = Ordinary Least Squares; β^* denotes standardized coefficients; SE = standard error; FE = fixed effects (within estimator). The fixed-effects model is reported as a robustness check rather than as definitive evidence, given the small cross-sectional dimension (N = 4 countries). The symbol "–" denotes no applicable value for this column.

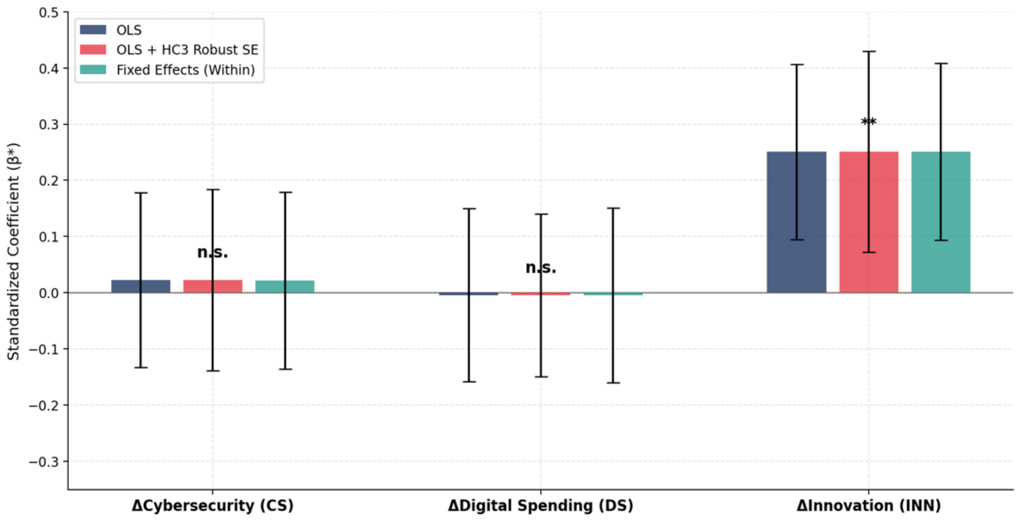

Figure 6. Standardized coefficient comparison across estimation methods (H4)
Note: Error bars represent 95% confidence intervals. OLS = Ordinary Least Squares; HC3 = heteroskedasticity-consistent covariance matrix estimator type 3; n.s. = not significant. **: $p < 0.01$.

Regression Diagnostic Tests

The H4 model passed diagnostic tests for the regression assumptions. For instance, the residuals are normally distributed (Shapiro-Wilk $W \approx 0.99$; $p > 0.05$) (Figure 7), and the Breusch-Pagan test shows no evidence of heteroskedasticity (which confirms that HC3 corrections are more conservative than corrective). The Durbin-Watson statistic of approximately 3.15 indicates that any autocorrelation issues originate from the MA(1) artifact introduced by first differencing; thus, using first-differenced OLS models will typically have a DW statistic exceeding 2.0 due to differencing having added a negative MA(1) contribution in the error term's composite error term (Wooldridge, 2010). Similar results are also obtained when estimating the original error term: within-country autocorrelation residuals provide evidence of such a phenomenon ($\rho \approx -0.5$). Therefore, these diagnostic results support the validity of applying OLS to the first-differenced specification, with the observed autocorrelation attributable to the mechanical effect of first differencing rather than to model misspecification.

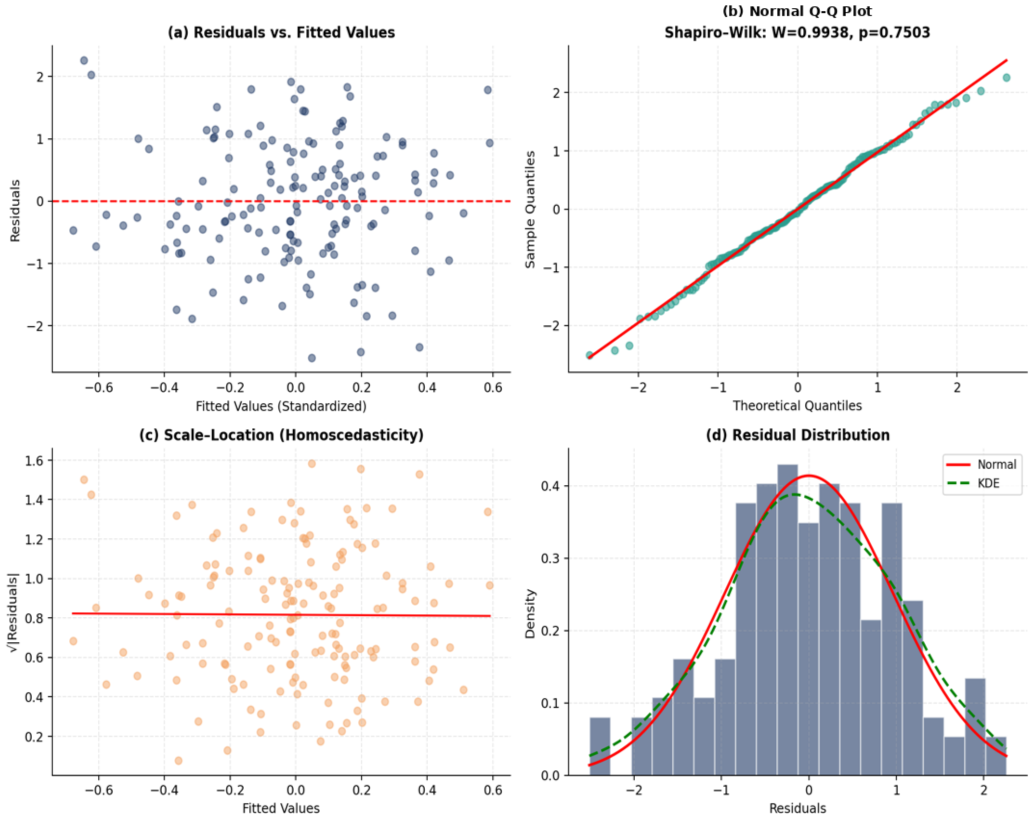

Figure 7. Regression diagnostic plots for H4 first-differenced model: (a) residuals vs. fitted; (b) normal Q–Q; (c) scale–location; (d) residual distribution with KDE and normal overlay

Note: KDE = Kernel Density Estimate; Q–Q = quantile-quantile plot.

5. Research Limitations and Future Directions

The research conducted has a few limitations. The first limitation pertains to the selection of the four countries in question, which makes it impossible to apply the findings to other countries currently developing their digital economies through different institutional frameworks. It is recommended to conduct comparative research in the future with a larger sample and including other countries, such as those from ASEAN, MENA, or Sub-Saharan Africa. The second limitation is the linear interpolation of annual GII scores into quarterly data, which leads to smoothing of the data and decreases its variability. This means that it is crucial to use original quarterly indicators of innovation, such as patent applications, R&D expenditures data, or AI usage indexes, when such information emerges on the market. The third limitation is the choice of using a first-differenced specification to achieve stationarity, which limits the identification to short-run contemporaneous relationships. Moreover, the absence of statistical significance for CS and digital expenditure does not imply a lack of economic impact; rather, it suggests the existence of a lag structure where investments require several years before an economic effect materializes.

The fourth limitation relates to the small cross-sectional dimensionality of the sample ($N = 4$), which limited the statistical power of using the fixed-effect estimator and, therefore, eliminated the possibility of using dynamic GMM (Arellano-Bond) or common-correlated effects (CCE) estimators (which require $N > 20$) in this study. Fifth, the GII is a composite index whose sub-components—innovation inputs vs. outputs—may have differential effects on growth, a distinction the aggregate score cannot capture. Sixth, the CS and digital-spending indicators are compiled from multiple sources (e.g., ITU and Access Partnership for CS; World Bank, OECD, and national budget reports for DS). Although the series were harmonized to common definitions and units and the analysis relies on standardized, first-differenced within-country variation to mitigate level-comparability concerns, some residual non-comparability across countries and sources cannot be entirely excluded and may attenuate the estimated effects of these two variables.

6. Conclusions

We studied how the amount of money spent on digital and cybersecurity investments and innovation could affect economic growth for the countries of Jordan, Saudi Arabia, Malaysia, and the UAE, using a first-differenced quarterly panel dataset and employing multiple estimation methods. Because we used a first-difference approach to conduct our analysis, we were able to resolve the issue of unit root non-stationarity and reduce our VIF values to negligible levels, thus enabling us to provide an unbiased estimate of the short-run coefficients.

Our major finding shows that only innovation could statistically predict short-run economic growth across the four countries we studied ($\beta^* \approx 0.25$; $p < 0.01$) which is consistent with the endogenous growth theory that states that technological change is the main source of productivity and growth. Although there is some theoretical justification for digital spending and cybersecurity investment impacting economic growth in the short run, these two variables do not achieve a statistically significant influence on economic growth, but this should be interpreted based on the Investment Lag Theory rather than represent a finding of economic irrelevance. We emphasize that, because the first-differenced specification identifies only contemporaneous short-run relationships, the non-significant coefficients on cybersecurity and digital spending should be read strictly as the absence of short-run predictive power in this specification and not as evidence that these investments are economically irrelevant over the longer term; their benefits are expected to accrue gradually through multi-year implementation, institutional trust, and infrastructure complementarities. The results of our Hausman test indicate the appropriate use of the fixed-effects specification in our estimation, while our robustness analysis (using OLS, HC3, and fixed-effects) consistently confirms our inferences about economic growth.

Specific policy implications of these findings include the following: (1) investment in innovation ecosystems should be made a priority since it provides quick returns in terms of developing a digital economy; (2) investments in cybersecurity and digital infrastructure should take into account multiyear implementation plans, with return on investment (ROI) approaches that go beyond one fiscal year; and (3) national digital systems that integrate innovation, cybersecurity, and investment in digital infrastructure are required to produce the complementarities observed in the literature.

7. Policy Recommendations

The priority for short-term growth is developing the innovation ecosystem. All four governments should consider prioritizing investment in R&D infrastructure, AI adoption programs, technology transfer agreements, and innovation hub establishment as there are no other significant predictors in the short-term model other than innovation ($\beta^* = 0.2508$).

Cybersecurity and digital spending programs may benefit from adopting a clearly defined ROI framework oriented toward the long term. Because their returns are subject to investment-lag effects, the evaluation of cybersecurity and digital infrastructure should be assessed over a 3–5-year time horizon, with budget allocations ideally tied to milestone-based outputs rather than expected to yield measurable short-run growth.

Digital expenditure should ideally follow protocols for strategic alignment. Given the absence of a significant short-run effect for digital spending, such expenditures may be more effective when strategically aligned so that adequate cybersecurity and innovation capacity is in place before growth benefits can be expected to materialize over the longer term.

Quarterly monitoring for digital economy indicators is valuable. A higher frequency of data collection (quarterly) illustrates the dynamics of the digital economy that are not visible when measured annually. The four nations may benefit from investing in high-frequency statistics on the digital economy to strengthen the evidence base of their policy-making processes.

Establish the ability for knowledge transfer across the four countries through exchange mechanisms. The four countries are committed to digital transformation and have complementary strengths that can enhance the success of their respective transformation efforts (UAE in cybersecurity, Malaysia in innovation, Saudi Arabia in digital spending scale, Jordan in institutional adaptation), creating natural conditions for bilateral and multilateral knowledge exchange programs.

Author Contributions

Conceptualization, J.Y.A. and Z.M.S.Z.; methodology, J.Y.A. and B.A.-S.; software, A.A.A.; validation, Z.M.S.Z., B.A.-S., and N.N.R.; formal analysis, J.Y.A. and A.A.A.; investigation, A.A.A. and N.N.R.; resources, J.Y.A.; data curation, A.A.A. and N.N.R.; writing—original draft preparation, J.Y.A.; writing—review and editing, Z.M.S.Z., B.A.-S., A.A.A., and N.N.R.; visualization, A.A.A.; supervision, Z.M.S.Z. and B.A.-S.; project administration, J.Y.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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